

Review of Activity Intelligence Algorithms for Higher-level Data Fusion

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1 Executive Summary

This report reviews algorithmic methods for extracting structured knowledge and predictive insight from semi-structured or tabular sources, focusing on the operational setting where these data arrive from reconnaissance platforms in real time. It covers three core layers:

1. Knowledge Base Generation from Tabular Data – Methods for converting tabular ISR outputs into semantically annotated knowledge graphs (KGs) using state-of-the-art entity linking and annotation techniques.
2. Semantic and Temporal Reasoning over KGs – Algorithms that use KGs to inform, constrain, or verify AI inference pipelines, including reasoning with large language models (LLMs) and forecasting future states.
3. Inference on Activities from Trajectories – Activity recognition models trained to map uncertain, sequential tracking data to activity categories, behaviors, and anticipated intent.

Hybrid systems that combine KG-based reasoning with contextual sensor fusion appear most promising, but they remain immature for theater deployment. Current algorithms require heavy task-specific tuning, significant compute, and lack clear benchmarks for operational readiness. In short: they are promising but not yet practical.

2 Overview of Activity Based Intelligence (ABI)

The volume, variety, and velocity of modern ISR data—particularly from airborne or remote reconnaissance platforms—has overwhelmed traditional analytic pipelines and prompted a shift toward Activity-Based Intelligence (ABI). ABI is a multi-source, multi-resolution framework that seeks to interpret the actions and intentions of people, vehicles, and systems by leveraging their patterns of motion, contextual signals, and historical behaviors [1]. It replaces static object-centric views with a temporal, behavior-driven model of intelligence.

Activity intelligence operates downstream of entity detection and tracking. It receives tabular trajectory data streams, often in noisy or incomplete formats, and must build semantic understanding from them. The mission goal is not merely to describe what is seen, but to infer why it is occurring and what may happen next. This requires algorithms capable of semantic enrichment, knowledge integration, and predictive modeling. An overview of the algorithms reviewed in this report can be found in table 1.

Algorithm	Citation ID	Year	Input Type
MantisTable	[11]	2019	Tabular Data, KG
DAGOBAB	[7, 22]	2019, 2021	Tabular Data, KG
TabEL	[3]	2015	Tabular Data, KG
ColNet	[9]	2018	Tabular Data (Text)
TabEAno	[43]	2020	Tabular Data, KG
TableLlama	[67]	2024	Tabular Data, KG
Table-GPT	[32]	2024	Tabular Data, KG
JAKET	[66]	2022	Text, KG
RoG	[39]	2024	Text, KG
CoK	[63]	2024	Text, KG
Dynamic KG Multi-Event Forecasting	[13]	2020	KG (Temporal)
Evolutional Representation Learning (Temporal KG)	[34]	2021	KG (Temporal)
Temporal Inductive Path Neural Network	[14]	2024	KG (Temporal)
CAVIAR	[2]	2019	Sensor Data, KG
Language of Actions	[46]	2025	Sensor Data
RAG	[31]	2021	Text, KG (Retrieval)
UniMS-RAG	[62]	2024	Text, Multisource Retrieval
RETRO	[6]	2022	Text, KG (Retrieval)
InstructRetro	[61]	2024	Text, KG (Retrieval + Instruction)
RECITE	[58]	2023	Text, Retrieval (Inference)
G-Retriever	[19]	2024	Text, Graph-based Retrieval

Table 1: Overview of reviewed algorithms with publication year and input data types.

2.1 Core Technical Challenges

ISR trajectories frequently arrive with incomplete or uncertain metadata. To enable meaningful semantic interpretation, methods for transforming such tabular data into KGs are presented, thereby supporting downstream reasoning over actions, contexts, and relationships.

This review focuses on three of four primary intelligence challenges:

1. Tabular Semantic Representation

How can sparse, irregular, and unstructured trajectory tables be reliably transformed into semantically rich formats such as KGs? Section 3 surveys state-of-the-art methods for addressing this problem.

2. Context-Enriched Reasoning

How can AI systems integrate external knowledge and historical patterns to improve inference? A prevalent strategy is to embed contextual information from KGs into prompts for a large language model (LLM). Section 4 examines common approaches to this integration.

3. Uncertainty and Temporal Integration

How to reason over data streams that evolve over time yet contain sensor noise, missing fields, and ambiguous behaviors? Section 4.2 explores methods for handling temporal KGs, while Section 5 introduces algorithms designed specifically for tracking scenarios.

4. Operational Interpretability and Modularity

How to design pipelines that are fast, auditable, edge-deployable, and adaptable for users with varied technical expertise? While understanding this question is fundamental for developing activity intelligence systems, it falls out of the scope of this report.

3 Generating a Knowledge Base from Tabular Data

The higher level data fusion task deals with post-processed raw data that can come in various formats. A common methodology for storing these data is in tabular format. To derive meaningful information from the tabular data, methods for deriving KGs or semantic webs from the tabular data have been developed. We define a KG the same as [20] which is “a graph of data intended to accumulate and convey knowledge of the real world, whose nodes represent entities and whose edges represent potentially different relations between these entities”, an example of which is shown in figure 1. A review on KGs and their applications can be found in [24].

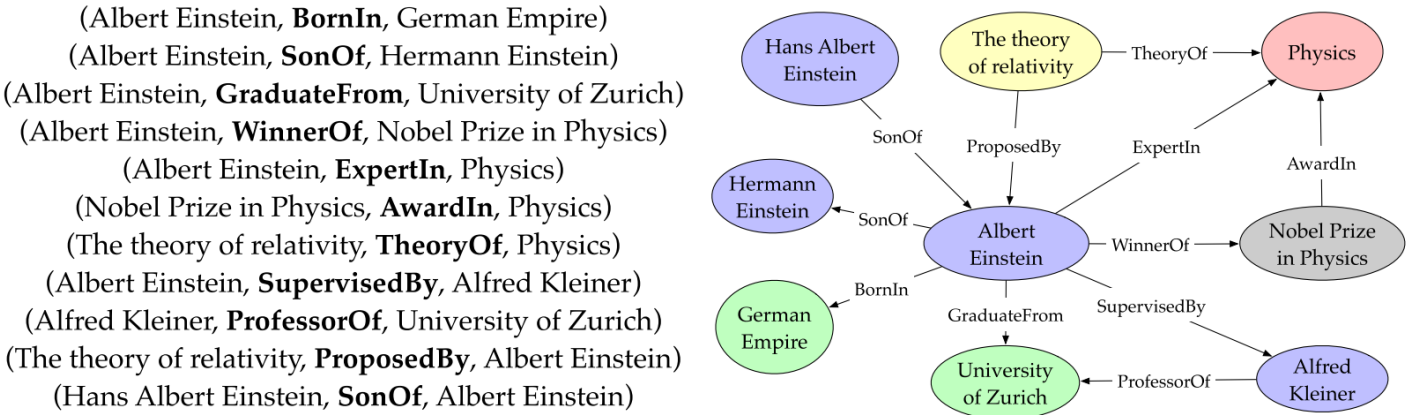


Figure 1: Example of a knowledge base of triplets (left) and its corresponding knowledge graph (right), taken from [24].

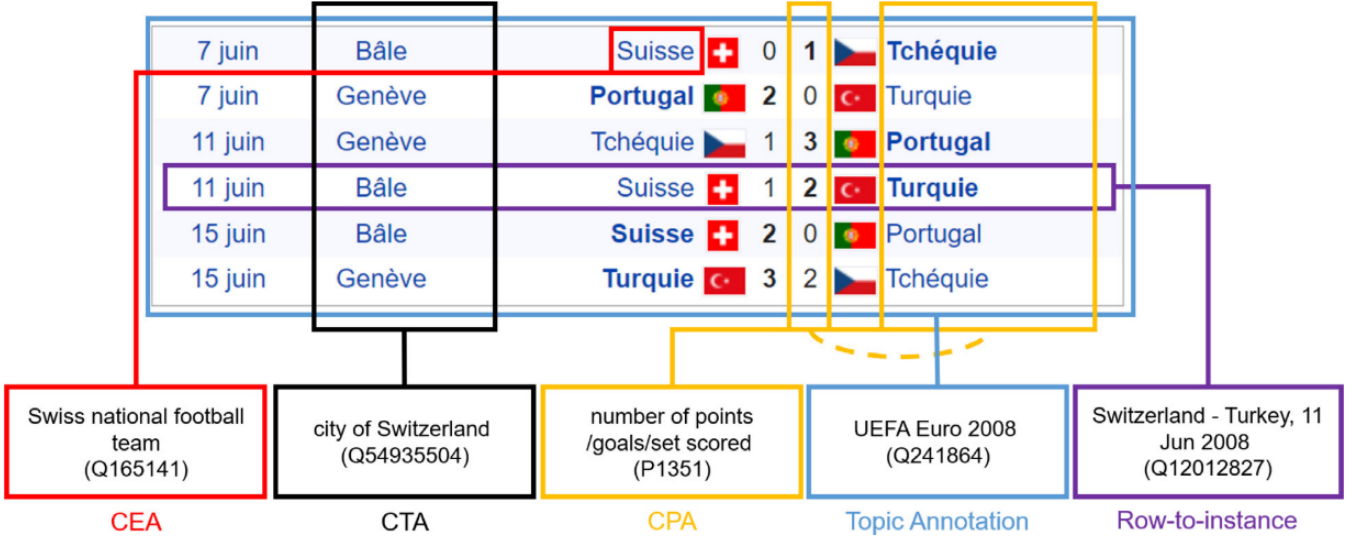


Figure 2: This figure describes the various annotation tasks that these algorithms are meant to solve, taken from [36].

A common strategy for constructing a representation graph is the semantic table interpretation (STI) method. Although technically sophisticated, most STI systems such as DAGOBAB and MantisTable are optimized for relatively clean and well-structured tables. In intelligence, surveillance, and reconnaissance (ISR) pipelines the tabular data usually arrive incomplete, irregular, or noisy. For this reason, benchmark performance results reported on controlled datasets like SemTab should be interpreted with caution, since they are unlikely to generalize to operational conditions.

At its core, the STI task involves aligning tabular data entries, including rows, columns, and cell items, with entities and relationships in an existing knowledge graph (KG). The SemTab challenge is a recurring evaluation that invites contributors to develop and test new methods for solving this mapping problem [18]. The main tasks defined in SemTab, which include cell-entity annotation, column-type annotation, column-property annotation, and table-level topic assignment, are summarized below and illustrated in figure 2.

1. **Cell-Entity Annotation (CEA)**: This task involves matching a cell with a KG entity. In figure 2 this is illustrated by Suisse being matched with the Swiss national football team, Suisse being the table entry and the Swiss national football team being the KG entity.
2. **Column-Type Annotation (CTA)**: This task involves matching a table column with a KG entity type. Shown in figure 2 by the column with the cities of Switzerland (Bale, Geneve) being matched with the appropriate KG entity.
3. **Columns-Property Annotation (CPA)**: Aims to annotate a column pair of a relational table with a property. For example in figure 2, the 5th and 7th columns are assigned the relationship of “number of points scored”.
4. **Topic annotation**: Aims to map the entire table to an entity in the target KG.
5. **Row-to-Instance**: Attempts to map the entire row of the table to an entity in the target KG.

Dataset	CEA	CTA	CPA
SemTab 2021	DAGOBAB (F1: 0.974) DAGOBAB [7, 22]	DAGOBAB (F1: 0.99) DAGOBAB [7, 22]	MantisTable (F1: 0.965) [11, 12]
Limaye	TabEL (Acc: 0.894) [3]	T2K++ (F1: 0.88) [53]	MantisTable (F1: 0.84) [11, 12]
T2D	TabEAno (F1: 0.91) [43]	ColNet (F1: 0.976) [9]	T2K++ (F1: 0.91) [53]

Table 2: Best-performing systems per task for SemTab 2021, Limaye, and T2D datasets.

3.1 Heuristic Based Table Annotation Methods

Heuristic based table annotation approaches are those class of algorithms that utilize custom, hand-crafted metrics for mapping the table entities (columns, rows, row items) to KG entities (entities, relationships). Of all the algorithms reviewed in [24], those that demonstrated superior performance in the various annotation tasks (illustrated in figure 2) are selected for review herein.

MantisTable (2019) MantisTable [11] is an unsupervised system for STI that automatically annotates independent tables, even when no metadata such as captions, surrounding text, or column headers are available. Its main contribution is to provide a comprehensive end-to-end pipeline for semantic table interpretation using only the intrinsic structure and content of the table. Unlike many previous STI systems, MantisTable does not rely on web context, personalized knowledge graphs, or pre-annotated training data. Instead, it performs all major STI tasks (entity linking, column typing, relation detection, and datatype annotation) in a unified and fully automatic way. This makes it applicable to independent tables in diverse domains without requiring external metadata or supervision.

DAGOBAB (2019, 2021) DAGOBAB [7,22] addresses the joint tasks of CEA, CPA, and CTA for table annotation. Its key contribution lies in extensive preprocessing and the use of a factor graph model to capture dependencies between entities, column types, and relations.

1. **Preprocessing:** Given a raw table, DAGOBAB infers its orientation, detects header rows, identifies key columns, and infers the data type of each column (e.g., distance, speed, email, IP address). These steps provide structural cues before linking table content to the KG.
2. **Candidate generation:** For each cell string s , a set of entity candidates C is retrieved from the KG based on string similarity and lexical matching. Similarly, each column is assigned a set of candidate semantic types, and each column pair is assigned candidate relations.
3. **Joint disambiguation:** The three tasks, are modeled jointly using a *factor graph*, where variable nodes correspond to candidate entities, column types, and relations, and factor nodes encode compatibility constraints among them. Factors include (i) entity–type compatibility, (ii) entity–relation compatibility, and (iii) type–relation compatibility. Probabilistic inference (e.g., belief propagation) is then applied to compute the most consistent assignment of entities, types, and relations across the table.

The result is a coherent annotation in which cell-level entities, column-level types, and inter-column relations reinforce one another. Compared to stepwise approaches, DAGOBAB’s joint factor graph formulation reduces error propagation and improves robustness in the absence of surrounding context.

3.2 Machine Learning Based Approaches

These methods rely less on hand crafted metrics for increased performance, and rely more on information extracted from the tables directly utilizing machine learning approaches like logistic regression and convolutional neural networks (CNNs).

TabEL (2015) TabEL [3] follows a three-step process for mapping a potential mention m to its corresponding entity e in the KG, where a potential mention is a phrase of text found in a table whose corresponding entity in the KG is unknown.

1. **Mention identification:** Traverse the table and identify all potential mentions.
2. **Entity candidate generation:** For each potential mention m , identify a set of possible KG candidates C .
3. **Disambiguation:** For each mention m , select an entity $e \in C$ as the most likely match.

Before starting the mapping process, a probability distribution $P(e | s)$ is computed, where s is a surface form string observed in a table cell. $P(e | s)$ represents the probability that entity e is the correct entity given the string s , typically estimated from hyperlink statistics or mention–entity co-occurrence in the source KG. Mention identification is then performed by iterating over each cell in the table and extracting the strings $s \in S$ that have a nonzero probability $P(e | s)$. Candidate generation is carried out similarly: for each $s \in S$, the candidate set C contains all entities e such that $P(e | s) > 0$.

The final step of disambiguation employs the *Iterative Classification Algorithm* (ICA) [38]. ICA treats the disambiguation problem as a collective classification task on a *factor graph*, where nodes correspond to mentions and candidate entities, and edges encode relational features such as row- or column-level coherence between entity assignments. A local classifier (e.g., logistic regression) provides initial predictions using lexical features, and ICA iteratively refines these predictions by incorporating relational features from neighboring nodes until convergence. This approach allows TabEL to enforce coherence among entity assignments across the entire table, rather than resolving mentions independently.

ColNet (2018) ColNet [9] addresses the task of CTA by training a CNN-based classifier on column data.

1. **Embedding:** Each cell value in a column is tokenized, i.e. words are mapped into a continuous vector space using word2vec embeddings [40]. This produces a real-valued matrix representation for each column.
2. **Classification:** The embedded column representation is passed through a CNN that extracts semantic patterns from the text and outputs a probability distribution over possible column types.

3. **Label assignment:** The most probable type is selected as the column annotation, providing a mapping between the column and its semantic class in the KG.

Unlike heuristic-based methods that rely on lookup statistics or handcrafted similarity functions, ColNet leverages distributional semantics to capture deeper contextual meaning in column content. However, the use of a CNN requires more computational resources during both training and inference. Moreover, ColNet’s architecture is based on early neural models, and more recent approaches using transformers or pretrained language models could significantly improve performance.

3.3 LLM Based Table Annotation

These methods utilize the power of modern LLMs to accomplish the STI task.

TableLlama (2024) TableLlama [67] introduces a generalist LLM-based approach for table annotation and related tasks through large-scale instruction tuning. Its methodology consists of three main components.

1. **Instruction tuning:** A new dataset, TableInstruct, was constructed with over 2.6M instances from 1.24M real tables, spanning 14 datasets and 11 task types. Each instance pairs a table with a natural language instruction and expected output, enabling multitask learning.
2. **Model fine-tuning:** TableLlama builds on Llama 2 (7B) [60] and is fine-tuned using LongLoRA [10] to handle table inputs up to 8,192 tokens. This allows effective processing of large or complex tables.
3. **Unified prompting:** At inference time, inputs follow a standard template of Instruction, Table Input, optional Question, and Response, which supports consistent application across tasks.

TableLlama was evaluated on both in-domain and out-of-domain datasets, where it achieved performance on par with or better than specialized task-specific models. Its key contribution is shifting from heuristic or pipeline-based methods toward a single LLM capable of generalizing across diverse domains and table tasks.

Table-GPT (2024) Table-GPT [32] proposes *table-tuning*: continuing to train an LLM on diverse table tasks synthesized from real tables to strengthen table understanding.

1. **Table-tuning paradigm:** Instead of prompt engineering, the model’s weights are updated once (not task-specific) using a large corpus of synthesized table tasks built from real tables. This improves general table reasoning and robustness to unseen tasks.
2. **Multi-level augmentation:** Four complementary augmentations mitigate overfitting and boost generalization: task-level, table-level, instruction-level, and completion-level. These diversify task formats, table structures/values, prompts, and outputs during training.
3. **Foundation model role:** The resulting model excels in zero-/few-shot table tasks and serves as a stronger starting point than vanilla GPT for downstream single-task optimization (fine-tuning or prompt design).
4. **Released resources:** Code, training data, and an extensive evaluation benchmark are publicly released to support reproduction and further research.

Overall, Table-GPT shifts from heuristic or per-task tuning toward a unified fine-tuned LLM specialized for diverse table tasks via synthesized data and principled augmentation.

3.4 Conclusion

Between heuristic-based, machine learning, and LLM-based approaches for table annotation, there are notable trade-offs. Heuristic-based methods are generally lightweight, requiring far less computational power and resources than LLM-based techniques, but require hand-crafted metrics that may need to be continuously updated to maintain performance. Machine learning based approaches take a more unsupervised approach, inferring the information directly from the tables and require less manual involvement. Lastly, LLM-based methods tend to achieve higher accuracy and stronger generalization across diverse datasets, but there is still a possibility of falsifying information.

4 Semantic Reasoning Algorithms

After KGs have been extracted from the data, the task evolves into performing inference on these KGs given new information. Whether that be raw text information, or new entities and relationships in the graph. Prior to the recent success of large language models in semantic inference over knowledge graphs, early approaches relied on probabilistic frame-based systems that extended Bayesian networks to relational settings [28], statistical relational models such as Markov Logic Networks [52], and early embedding-based methods like TransE [5], which collectively laid much of the groundwork for modern KG reasoning. However, given the potential reasoning capabilities recently demonstrated by LLMs [21], our review focuses exclusively on LLM-based approaches to semantic reasoning.

4.1 Reasoning with LLMs using Knowledge Graphs

By leveraging the structured relationships encoded in our KGs, LLMs inferences can be grounded in factual context, reducing hallucinations (incorrect inferences from large language models) and ensuring consistency. These KGs serve both as a source of evidence for the LLM and as a post hoc verifier of its outputs, creating interpretable reasoning chains.

While a wide variety of methods exist for integrating KGs into LLM reasoning, the following three algorithms were selected because they illustrate three distinct integration strategies:

- Joint learning of text and KG representations in a unified model (JAKET).
- Constrained reasoning that forces LLMs to follow valid KG paths (Reasoning on Graphs).
- Iterative fact retrieval and refinement to dynamically guide reasoning (Chain of Knowledge).

Together, they provide representative examples of how KGs can enhance LLM reasoning, from the data integration stage, through structured guidance during inference, to adaptive retrieval in the reasoning loop.

JAKET, Joint Pre-training of Knowledge Graphs and Language Understanding (2022) JAKET [66] is a framework for training a pre-trained large language model in conjunction with a graph neural network, in this case a graph attention network (GAT), to be able to create a KG from raw text. They use English Wikipedia as the text corpus, and Wikidata as the KG. With these, they are able to train their model to learn entity categories and relation prediction.

While JAKET is primarily aimed at building and enriching KGs from unstructured text, the underlying idea, jointly modeling structured and unstructured data, can be adapted to domains such as tracking and activity intelligence when sensor-derived events are first translated into structured tabular representations.

- **Pros:** Not a complex system, the paper clearly outlines the implementation details, and its performance is state of the art for the year 2022.
- **Cons:** Mainly used for raw text data, though integration with tabular data pipelines seems feasible.

Reasoning on Graphs (RoG) (2024) RoG [39] constrains LLM inference to valid relation paths extracted from a KG, using a two-stage planning and retrieval–reasoning pipeline to guide the model along minimal-divergence paths. It achieves state-of-the-art KG-grounded question answering by first generating abstract relation sequences and then retrieving concrete entity walks to inform the answer generation process.

- **Pros:** Interpretable reasoning chains grounded in actual KG structure; large accuracy gains on standard KG question answering (KGQA) benchmarks; plug-and-play planning module that boosts off-the-shelf LLMs without retraining.
- **Cons:** Requires comprehensive entity linking and KG coverage (performance degrades if facts are missing); retrieval over large graphs can add latency; two-stage fine-tuning is more complex and data-hungry than end-to-end approaches.

Chain of Knowledge (CoK) (2024) CoK [63] interleaves LLM reasoning with dynamic fact retrieval by fine-tuning on exemplars that illustrate correct KG and corpus usage; at inference, it alternates between generating candidate facts (triples or passages) and validating them against structured and unstructured sources before consolidating the final answer. This “generate, retrieve, refine” loop yields more factual, self-verified chains of thought.

- **Pros:** Substantial reduction in hallucinations and improved factual accuracy over vanilla Chain-of-Thought; seamless access to heterogeneous knowledge sources via an adaptive query generator; modular design lets you swap out the LLM or query engine independently.
- **Cons:** Heavily dependent on the quality and currency of external knowledge bases; per-step query generation and execution incur nontrivial latency and engineering overhead; still vulnerable to downstream LLM reasoning failures even when facts are correct.

4.2 Temporal Reasoning over Knowledge Graphs

This class of algorithms aims to solve the temporal reasoning over KGs problem. In a **static KG**, entities and relationships remain fixed over time. For example: (Paris, capital of, France) is a fact that does not change. In a **temporal KG**, facts are time-dependent. The tuple also includes the period when the statement holds true, for example: (Obama, president of, USA, 2009-2017) or (airplane 1, flying to, JFK, Jan 1, 0900 – Jan 1, 1300).

Reasoning over such graphs encompasses tasks such as forecasting how relationships and entities will evolve (predicting when a statement will no longer be true or when a new relationship will appear) [13], question answering using temporal KG facts to respond to queries [34, 55], and real-time updates to adjust the graph as new events occur [14]. To address these challenges, a variety of algorithmic approaches have emerged, including event forecasting models that leverage historical patterns in the KG to predict future events [13], evolutionary representation learning models that efficiently update KG embeddings as entities and relations change [34], and path-based reasoning models that exploit temporal paths through the graph to capture both structural and time-dependent patterns [14].

The following three algorithms are chosen not as the only solutions in their categories, but as representative milestones that illustrate the diversity of approaches to temporal reasoning on knowledge graphs. They (i) are recognizable contributions in the literature, (ii) span different methodological categories, and (iii) illustrate the trajectory of progress from earlier forecasting models to more advanced path-based reasoning frameworks.

Dynamic Knowledge Graph based Multi-Event Forecasting (2020) Multi-event forecasting [13] was one of the first works to formalize the problem of predicting multiple future events from historical context in a temporal KG. It demonstrated how to represent entities and relationships at a given time and use them to infer probabilities of future events. Other forecasting approaches exist (e.g., RNN-based [25], reinforcement learning [57]), but this work remains a well-cited reference point.

- **Pros:** A novel method for using historical context to predict future events.
- **Cons:** Domain specific in practice, with limited usability outside political-event prediction.

Temporal Knowledge Graph Reasoning Based on Evolutional Representation Learning (2021) This model [34] exemplifies temporal representation learning by leveraging past known facts to predict future relations or entities. For instance, given historical triples at times $t-1$ and t , the model predicts likely facts at $t+1$. It was notable for its computational efficiency and ability to capture evolving temporal dynamics.

- **Pros:** Handles evolving temporal data with high runtime efficiency compared to state-of-the-art in 2021.
- **Cons:** Performance degrades when reasoning over large subsets of entities or relations.

Temporal Inductive Path Neural Network (2024) The Temporal Inductive Path Neural Network [14] represents recent advances in path-based reasoning. It introduces historical temporal graphs, which merge all timesteps into a unified structure, enabling more accurate and efficient inference.

- **Pros:** Achieves improved accuracy and efficiency over prior path-based methods.
- **Cons:** Complex architecture requiring significant engineering overhead.

5 Inference using tracking information

Research directly applying LLMs to raw tracking data remains minimal, and current evidence suggests that they cannot yet handle sparse, irregular, or adversarially degraded trajectory streams reliably. A more viable near-term strategy may be to adapt techniques for temporally evolving knowledge graphs, as discussed in Section 4, to tracking information. In practice, most existing algorithms focus on predicting and classifying actions from sensor and contextual data streams. These models typically take as input inertial and location signals, sometimes augmented with contextual metadata such as location type or time-of-day. The most widely studied setting for these techniques is the Human Activity Recognition problem, where the goal is to identify actions such as walking, running, driving, or sitting. This popularity stems from:

- The prevalence of personal devices with embedded sensors (accelerometers, gyroscopes, GPS, etc.), enabling easy collection of large datasets.
- Commercial and practical interest in applications for health monitoring, fitness tracking, and personalized recommendations.

Although many existing approaches target human activities, the same principles, processing time-series sensor data, modeling sequential patterns, and incorporating context, can be adapted to other domains.

CAVIAR (2019) The Context-aware ActiVe and Incremental Activity Recognition (CAVIAR) [2] system predicts the activity a user is performing by using the inertial sensors on a user’s wearables as well as external knowledge bases which provide context.

For example, inertial sensor readings (e.g. timestamp, sensor ID, velocity in x , y , and z directions) are first used to produce a list of possible activities. Then, contextual information such as semantic location (e.g., “at the gym”) or time of day is used to filter out activities that are unlikely in the current situation.

- **Pros:** Increased accuracy and model confidence over methods without context or using context as features. Ability to maintain high accuracy across activities that are and are not context dependent.
- **Cons:** Model accuracy is dependent on the accuracy of each knowledge source in the knowledge base. Semi-supervised model requires a training dataset that maps sensor readings to actions for training.

Language of Actions (2025) The Language of Actions [46] approach takes inspiration from how we interpret sequences in natural language. Instead of viewing sensor data as continuous numeric streams, it represents them as sequences of discrete “moves” similar to words in a sentence.

Each activity is modeled as a distinct “language” learned from examples of that activity’s movement patterns. At inference time, the system compares incoming sensor sequences against these learned activity models, selecting the most likely one. Because the method models the structure of activities explicitly, it can also forecast the next expected movement, enabling real-time prediction.

- **Pros:** Generative and interpretable, models can be examined like a vocabulary; adding new activities does not require retraining the entire system; achieves high accuracy even with a single sensor; naturally supports next move forecasting.
- **Cons:** Discretizing the data may lose fine detail; model complexity grows with the size of the learned “vocabulary”, which can limit scalability for richer sensor setups.

5.1 Probabilistic Baselines and Learning-based Extensions

Research on utilizing large language models (LLMs) directly for extracting semantic reasoning from tracking data remains minimal. Current literature provides little evidence that LLMs can effectively handle raw trajectory streams, particularly when data are sparse, irregular, or adversarially degraded. As a result, most practical work in this area relies on more traditional probabilistic and sequential models.

Classic approaches such as Kalman filters [26], Interacting Multiple Models (IMM) [4], Hidden Markov Models (HMMs) [49], and Conditional Random Fields (CRFs) [29] remain the operational backbone for trajectory inference. These models are valued for their ability to run in real time, handle uncertainty explicitly, and function on resource-limited platforms. While they may not achieve the expressive power of modern neural or LLM-based approaches, they are reliable under ISR conditions where compute, bandwidth, and training data are often limited.

The newer approaches, such as CAVIAR and the Language of Actions, demonstrate how context-aware or generative frameworks can extend beyond traditional filtering by incorporating semantic cues or treating motion as a structured “language.” These innovations point toward future hybrid systems that could blend the interpretability and efficiency of probabilistic filters with the adaptability of machine learning.

A critical gap in this field is the lack of benchmarking under realistic ISR conditions. Most activity recognition datasets used in research assume well-labeled human activity data collected in controlled environments. In contrast, ISR data often arrive with missing metadata, noisy sensor streams, or deliberate deception. Without head-to-head evaluations that compare probabilistic baselines against machine learning and LLM-inspired methods under these constraints, it is difficult to assess which approaches are truly viable for operational deployment.

5.2 Toward Semantically Integrated Tracking

A limitation of most current tracking inference methods is their narrow focus on numerical trajectories or raw sensor readings [54]. While probabilistic filters and sequence models are effective at handling uncertainty, they do not inherently capture the semantic meaning of observed actions. In ISR environments, understanding whether an entity is “loitering near a checkpoint” or “moving in convoy formation” may be more valuable than classifying its speed or direction alone.

Recent work points to the potential of coupling trajectory inference with semantic enrichment [68]. This includes linking tracked entities to knowledge graphs, incorporating contextual metadata such as location type or time-of-day, and embedding symbolic constraints into learning-based models. Such integration enables low-level movement data to be reinterpreted as higher-level activities, allowing systems to reason about intent and anticipate future behavior, rather than simply projecting motion.

The challenge is that these semantics-aware approaches demand richer contextual information and often require fusion across heterogeneous data sources. They also raise questions about computational feasibility at the edge and the robustness

of semantic mappings under adversarial or noisy conditions. Nevertheless, the promise of integrated semantics is clear: they can transform tracking from a descriptive task into an explanatory one.

By retrieving, fine-tuning, or pre-training with domain-specific knowledge, LLMs can serve as the semantic layer that interprets and enriches raw trajectory data, bridging the gap between traditional tracking pipelines and higher-level activity intelligence [47].

6 Embedding new Context into LLMs

For LLMs to accurately infer about an environment, added context may be necessary for it to understand and reason correctly. This is commonly known as retrieval augmented generation (RAG). A survey on the evolution of these algorithms can be found in [16], which highlights three main methodologies.

1. **Fine-tuning:** This involves training the model to not only provide the expected output from a prompt, but also perform a query for information that will help answer the prompt better. This training happens post LLM training.
2. **Pre-training:** Similar to fine-tuning, however this involves training the model from scratch to both answer prompts and query for information that will allow it to answer prompts better.
3. **Inference:** The most commonly used and easiest to implement method, it involves using a trained LLM, and then concatenating relevant information to the users prompt to enable a better response.

For each type of context embedding method, we introduce the foundational paper along with the state of the art paper in the field.

6.1 Fine-tuning Methods

Retrieval-Augmented Generation (RAG) (2021) The RAG framework [31] was the first to demonstrate how integrating a retrieval component into an LLM pipeline could substantially improve performance on knowledge-intensive tasks. It combines a dense retriever with a sequence-to-sequence model, allowing the system to fetch relevant passages from a knowledge corpus and then condition its generative output on those passages. This fine-tuning approach provided a principled way to align retrieval with generation, reducing hallucinations and enabling stronger factual grounding.

UniMS-RAG (2024) Building on RAG, UniMS-RAG [62] represents the state of the art in fine-tuned retrieval-augmented methods. Unlike the original RAG, which primarily focused on single-source retrieval, UniMS-RAG integrates multiple retrieval sources (structured databases, unstructured text, and multimodal data) into a unified retrieval-augmented generation pipeline. The fine-tuning process is optimized to allow the model to dynamically select and weight sources based on task demands. This improves adaptability across domains, enabling more accurate responses in open-domain settings and specialized applications alike.

6.2 Pre-training Methods

RETRO (2022) RETRO (Retrieval-Enhanced Transformer) [6] introduced retrieval directly into the pre-training phase of a large language model. Rather than fine-tuning on retrieval tasks post hoc, RETRO trained the model from the ground up to interleave token prediction with retrieval of semantically similar contexts from a massive external corpus. This demonstrated that retrieval-augmented pre-training could achieve performance comparable to models 25 times larger, thereby reducing computational requirements while maintaining quality.

InstructRetro (2024) InstructRetro [61] extended RETRO by combining retrieval-augmented pre-training with instruction tuning. The model is not only trained to retrieve relevant content but also optimized to follow natural language instructions, aligning it better with real-world user queries. This hybridization of retrieval-pretraining and instruction alignment represents the current frontier, improving controllability, factual grounding, and user intent alignment simultaneously.

6.3 Inference Methods

RECITE (2023) RECITE [58] exemplifies inference-time retrieval augmentation by introducing a method that encourages models to explicitly cite external passages during response generation. Rather than simply concatenating retrieved passages to a prompt, RECITE integrates them in a structured way, guiding the model to both ground its answers in retrieved evidence and to present references in its output. This reduced hallucinations and made model reasoning more interpretable.

G-Retriever (2024) G-Retriever [19] represents the state-of-the-art in inference-based retrieval augmentation. It introduces a graph-based retriever that models relationships between documents, enabling more contextually coherent and multi-hop retrieval compared to earlier dense or sparse methods. By enriching prompt concatenation with relationally aware context, G-Retriever enhances reasoning across documents, making it particularly effective for complex queries that require synthesis across multiple sources.

6.4 Comparative Summary

Across these approaches, a clear progression can be observed. Fine-tuning methods have evolved from basic retrieval integration (RAG) to multi-source adaptive retrieval (UniMS-RAG), enhancing flexibility and domain robustness. Pre-training approaches demonstrate that embedding retrieval into the model’s training process (RETRO) can significantly reduce scaling demands, with recent work (InstructRetro) coupling retrieval with instruction tuning to better align with user intent. Inference methods, while the easiest to implement, have advanced from explicit citation mechanisms (RECITE) to sophisticated graph-based retrieval strategies (G-Retriever) that support multi-hop reasoning. These methods demonstrate the feasibility of modifying LLMs for specific domains, greatly increasing their usability and effectiveness.

7 Conclusion

Despite rapid progress in semantic AI and knowledge-driven reasoning, practical deployment in real-time ISR environments remains limited. Classical heuristic and probabilistic models continue to offer reliable performance under constrained conditions, but they lack the adaptability, semantic depth, and foresight needed for higher-level activity intelligence.

In contrast, recent advances in large language model applications show a clear trajectory toward superior performance. Heuristic methods such as DAGOBAB provide only a stopgap: low-latency and functional, but easily outperformed by LLM-based table annotation and reasoning systems when sufficient compute is available. Hybrid approaches like CoK and RoG demonstrate that LLMs can be combined with knowledge graphs to deliver interpretable, auditable, and context-aware reasoning, attributes essential for mission-critical trust. Moreover, trajectory inference methods inspired by LLM architectures—such as the Language of Actions—are beginning to surpass probabilistic baselines by enabling next-move forecasting, structured sequence modeling, and real-time intent recognition.

The strongest future direction lies in embedding mission-critical context into LLMs. Retrieval-augmented generation, fine-tuning with ISR-specific corpora, and hybrid KG-LLM pipelines all point to models that can adapt on the fly, reason across heterogeneous data streams, and maintain explanatory transparency. While challenges remain around compute, bandwidth, and adversarial robustness, the evidence suggests that LLM-driven solutions will increasingly define the state of the art in semantic reasoning and activity intelligence.

8 Appendix A: Neural Architecture Reference

To support a clear understanding of the methods reviewed in this report, a brief overview of core deep learning architectures that underpin many of the algorithms is provided. These foundational concepts, such as multi-layer perceptrons (MLPs) and CNNs, form the basis for various sensor fusion, classification, and reasoning tasks. A detailed introduction to these architectures is provided in [45], which outlines their structure and function in the context of signal processing and pattern recognition.

8.1 Multi-Layer Perceptrons (MLP)

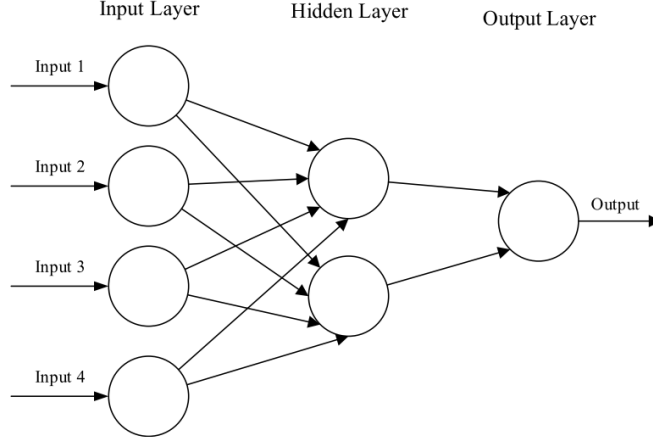


Figure 3: An example of a feed forward neural network (FFNN) also known as a multi-layer perceptron (MLP).

8.1.1 MLP Example

Consider a multilayer perceptron (MLP) with:

- Input vector $\mathbf{x} \in \mathbb{R}^4$
- Hidden layer with 2 neurons using ReLU activation
- Output layer with a single neuron using sigmoid activation

Let the weight matrix and bias vector of the hidden layer be

$$\mathbf{W}_1 \in \mathbb{R}^{2 \times 4}, \quad \mathbf{b}_1 \in \mathbb{R}^2$$

The pre-activation (linear) output of the hidden layer is given by:

$$\mathbf{z}_1 = \mathbf{W}_1 \mathbf{x} + \mathbf{b}_1$$

Apply the ReLU activation function element-wise:

$$\mathbf{a}_1 = \text{ReLU}(\mathbf{z}_1)$$

where

$$\text{ReLU}(z) = \max(0, z)$$

Let the weight vector and bias scalar of the output layer be

$$\mathbf{W}_2 \in \mathbb{R}^{1 \times 2}, \quad b_2 \in \mathbb{R}$$

Then the pre-activation output of the output layer is:

$$z_2 = \mathbf{W}_2 \mathbf{a}_1 + b_2$$

Applying the sigmoid activation function:

$$\hat{y} = \sigma(z_2) = \frac{1}{1 + e^{-z_2}}$$

The final prediction $\hat{y} \in (0, 1)$ represents the probability of the input belonging to class 1. The loss is then computed between the predicted and expected outputs, and the weights of the model are updated with respect to that loss via gradient based methods.

8.2 Autoencoder

Consider a simple autoencoder with:

- Input vector $\mathbf{x} \in \mathbb{R}^4$
- Encoder layer compresses to 2-dimensional latent space using ReLU activation
- Decoder layer reconstructs the original input using linear activation

Let the encoder have weight matrix and bias:

$$\mathbf{W}_e \in \mathbb{R}^{2 \times 4}, \quad \mathbf{b}_e \in \mathbb{R}^2$$

The latent (compressed) representation is:

$$\mathbf{z} = \text{ReLU}(\mathbf{W}_e \mathbf{x} + \mathbf{b}_e)$$

Let the decoder have weight matrix and bias:

$$\mathbf{W}_d \in \mathbb{R}^{4 \times 2}, \quad \mathbf{b}_d \in \mathbb{R}^4$$

The reconstructed output is:

$$\hat{\mathbf{x}} = \mathbf{W}_d \mathbf{z} + \mathbf{b}_d$$

The goal of training is to minimize the reconstruction error between the input and the output, typically using the mean squared error (MSE) loss:

$$\mathcal{L} = \|\mathbf{x} - \hat{\mathbf{x}}\|^2$$

During training, the weights $\mathbf{W}_e, \mathbf{b}_e, \mathbf{W}_d, \mathbf{b}_d$ are updated to reduce this loss, forcing the encoder to preserve the most important features of $\mathbf{x}$ in the latent vector $\mathbf{z}$.

8.3 Convolution Neural Network (CNN)

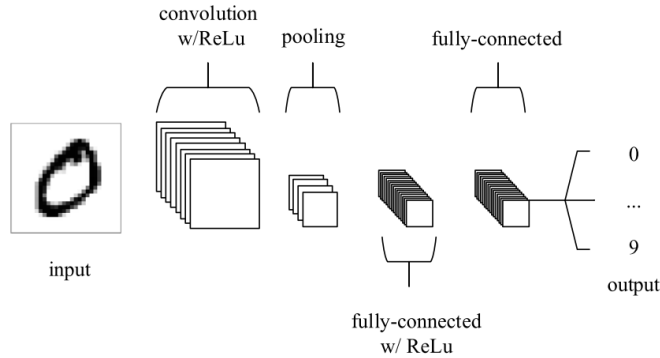


Figure 4: Unlike traditional methods that use fixed kernels like the Sobel filter, a CNN learns the optimal kernels directly from the data to produce the desired output.

8.3.1 CNN Example

Consider a simple CNN consisting of a single convolutional layer with the following components:

- Input image $\mathbf{X} \in \mathbb{R}^{3 \times 3}$
- One learnable filter (kernel) $\mathbf{K} \in \mathbb{R}^{2 \times 2}$
- Bias term $b \in \mathbb{R}$
- Stride $s = 1$, no padding
- ReLU activation function

The stride defines how many pixels the filter shifts as it moves across the input image during the convolution operation. Padding, on the other hand, is applied to preserve spatial dimensions when the filter does not naturally fit within the image boundaries. In this example, a 2×2 filter on a 3×3 input image with a stride of 1 and no padding is used. As a result, the filter slides across four valid positions in the input, producing an output feature map $\mathbf{Z} \in \mathbb{R}^{2 \times 2}$, where each value is computed by performing a dot product between the filter and the corresponding subregion of the input image.

Let the input image be:

$$\mathbf{X} = \begin{bmatrix} 1 & 2 & 0 \\ 3 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix} \quad \text{and filter:} \quad \mathbf{K} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}, \quad b = 0$$

The convolution outputs are:

$$\begin{aligned} z_{00} &= 1 \cdot 1 + 2 \cdot 0 + 3 \cdot (-1) + 1 \cdot 1 = 1 + 0 - 3 + 1 = -1 \\ z_{01} &= 2 \cdot 1 + 0 \cdot 0 + 1 \cdot (-1) + 1 \cdot 1 = 2 + 0 - 1 + 1 = 2 \\ z_{10} &= 3 \cdot 1 + 1 \cdot 0 + 0 \cdot (-1) + 2 \cdot 1 = 3 + 0 + 0 + 2 = 5 \\ z_{11} &= 1 \cdot 1 + 1 \cdot 0 + 2 \cdot (-1) + 2 \cdot 1 = 1 + 0 - 2 + 2 = 1 \end{aligned}$$

So the output before activation is:

$$\mathbf{Z} = \begin{bmatrix} -1 & 2 \\ 5 & 1 \end{bmatrix}$$

$$\mathbf{A} = \text{ReLU}(\mathbf{Z}) = \begin{bmatrix} \max(0, -1) & \max(0, 2) \\ \max(0, 5) & \max(0, 1) \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 5 & 1 \end{bmatrix}$$

The output feature map $\mathbf{A}$ represents the activations of the convolutional layer. This can then be flattened or passed to another convolutional or dense layer for further processing.

8.4 Attention

The attention mechanism is a central component of modern deep learning architectures, particularly in large language models such as ChatGPT. It provides a means of learning the relevance or similarity between vectorized entities, allowing the model to focus on the most informative parts of the input when making predictions. This mechanism enables efficient handling of sequential and multimodal data by dynamically weighting the contribution of different elements based on their contextual importance.

8.4.1 Attention Example

Given an input sequence of n tokens, each represented by a vector of dimension d , the attention mechanism computes a weighted representation of the values based on similarity between queries and keys.

The input matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ is linearly projected into three matrices:

$$\mathbf{Q} = \mathbf{XW}_Q, \quad \mathbf{K} = \mathbf{XW}_K, \quad \mathbf{V} = \mathbf{XW}_V$$

where:

- $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d \times d_k}$ are learnable weight matrices
- $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{n \times d_k}$ are the resulting query, key, and value matrices

Here n represents the amount of data, and d the dimensionality of the data, and d_k represents the arbitrarily chosen dimensionality of the output vectors.

Compute raw attention scores by taking the dot product of queries and keys:

$$\mathbf{S} = \mathbf{QK}^\top \in \mathbb{R}^{n \times n}$$

Then scale the scores:

$$\mathbf{S}_{\text{scaled}} = \frac{\mathbf{S}}{\sqrt{d_k}}$$

Apply softmax across each row to get attention weights:

$$\mathbf{A} = \text{softmax}(\mathbf{S}_{\text{scaled}}) \quad (\text{each row sums to 1})$$

Use the attention weights to compute the weighted sum of the values:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathbf{AV}$$

The output is a matrix of shape $\mathbb{R}^{n \times d_k}$, where each row represents a context-aware embedding of the corresponding input token.

9 Appendix B: Lower Level (0-3) machine learning based fusion techniques

9.1 Deep Learning in Data Fusion

Algorithm/Model	Sensor Modalities	Open Source	Usability
BEVFusion [37]	Camera + LiDAR	Yes	High
ImageBind [17]	Image, Audio, IMU, Text, Thermal, Depth	Yes	High
CRF-Net [44]	Radar + Camera	Yes	Medium
Attention Bottlenecks [42]	Multi-modal	Yes	Low
Perceiver [23]	Multi-modal	No	Low
SAM / SAM2 [27, 51]	Image + Video	Yes	High
PointPillars [30]	LiDAR	Yes	Medium
DiscoGraph [33]	Multi-agent LiDAR	Yes	Medium
JS3C-Net [65]	LiDAR (multi-frame)	Yes	Medium
PV-RCNN++ [56]	LiDAR	Yes	Medium
Vision Transformer (ViT) [15]	Image	Yes	Medium
RodNet [64]	Radar + Camera	Yes	Low
CenterFusion [41]	Radar + Camera	Yes	Low
Small Object Fusion [8]	Radar + Camera	No	Low
Change Detection (SAR + Camera) [35]	SAR + Optical Imagery	No	Low
CLIP [50]	Image + Text	Yes	High
VideoStreaming [48]	Video	Not Yet	Low

Table 3: Overview of multi-modal fusion models with sensor modalities, open source availability, and usability (based on security, ease of implementation, and existing development).

These neural network architectures contribute across different levels of data fusion. CNNs and multi-layer perceptrons (MLPs) are commonly employed at Level 0 and Level 1, where their strengths in low-level feature extraction and object detection are most effective. At higher levels, specifically Level 2 and beyond, attention mechanisms and transformer-based architectures become essential for modeling complex temporal dependencies and multimodal relationships, thereby enabling semantic reasoning and contextual understanding.

9.2 Level 0: PRE-PROCESSING Algorithms

These algorithms cover a variety of deep learning based methods for fusing and interpreting multiple types of data inputs. An excellent overview of deep learning based fusion models along with all publicly available datasets is provided by [59].

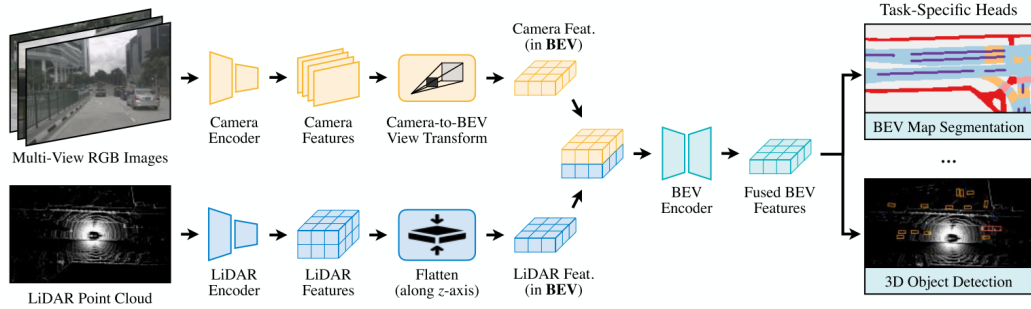


Figure 5: BEVFusion integrates camera and LiDAR data by projecting both modalities into a shared bird’s eye view (BEV) representation, allowing for enriched feature extraction and unified spatial understanding. This shared embedding enables multiple downstream tasks, such as semantic segmentation and object classification, to be performed more effectively. However, its utility is constrained by the requirement that both camera and LiDAR data originate from the same physical platform, such as an autonomous vehicle equipped with both sensors. This dependency limits the method’s generalizability in scenarios involving disjoint or heterogeneous sensor platforms, which are common in distributed surveillance and defense applications [37].

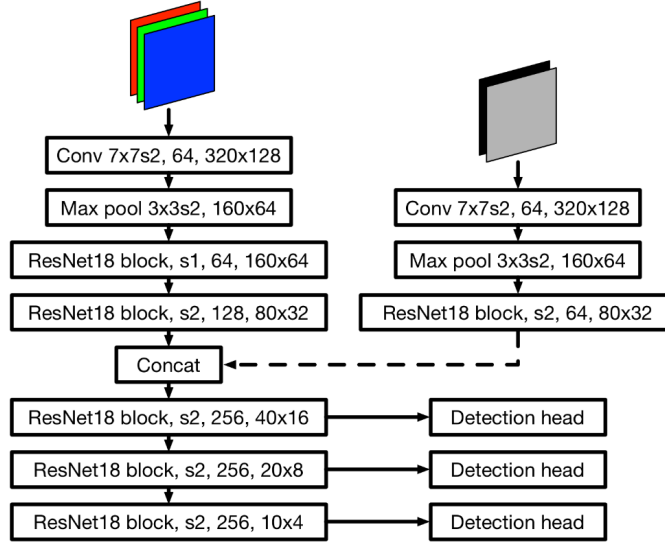


Figure 6: This study addresses the challenge of detecting small or visually ambiguous objects in modern computer vision systems. The proposed approach integrates radar measurements with camera imagery to enhance object detection performance. By leveraging radar’s robustness to lighting and weather variations, the model compensates for the limitations of camera-only perception. While the method effectively improves detection in constrained visual scenarios, its reliance on radar-camera pairing narrows its applicability to platforms equipped with both modalities. This sensor fusion strategy may not generalize well to scenarios where heterogeneous data sources are processed independently, as is often the case in higher-level data fusion contexts [8].

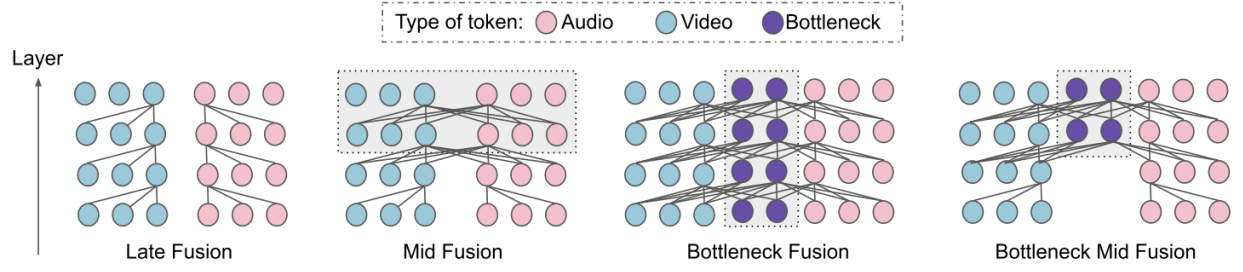


Figure 7: This study investigates the fusion of audio and video data to enhance multimodal classification tasks. By introducing an attention bottleneck mechanism, the model efficiently integrates these modalities before inference, reducing computational overhead. The authors demonstrate that this approach outperforms prior models that relied on a single modality. This fusion strategy is particularly valuable in scenarios where auditory information can disambiguate or reinforce visual context. However, the reliance on audio-visual data may limit its direct applicability in sensor networks where audio capture is unavailable or impractical [42].

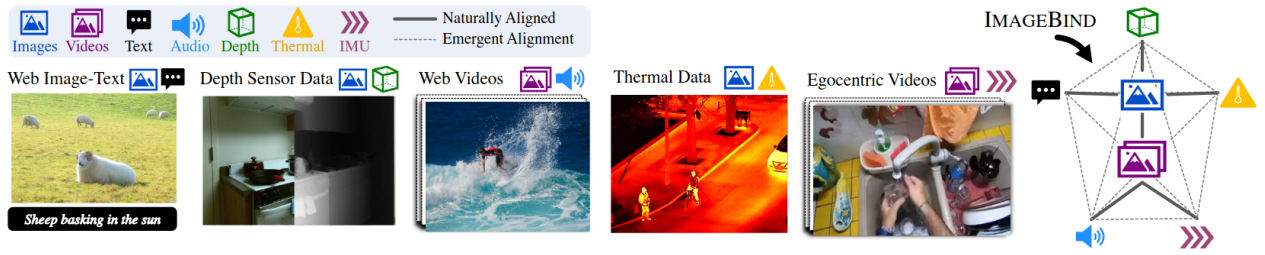


Figure 8: ImageBind is a multi-modal embedding framework designed to jointly represent six data modalities: images, text, audio, depth, thermal, and inertial measurement unit (IMU) data. The model is trained using paired image-modality inputs, enabling the learned embedding space to align information across modalities without requiring direct pairwise supervision for every combination. This shared embedding facilitates cross-modal retrieval and classification, often matching or surpassing the performance of models specialized for individual modalities. While the approach offers substantial flexibility in integrating heterogeneous data sources, its reliance on image-centric pairing during training may restrict its applicability in real-world data fusion contexts [17].

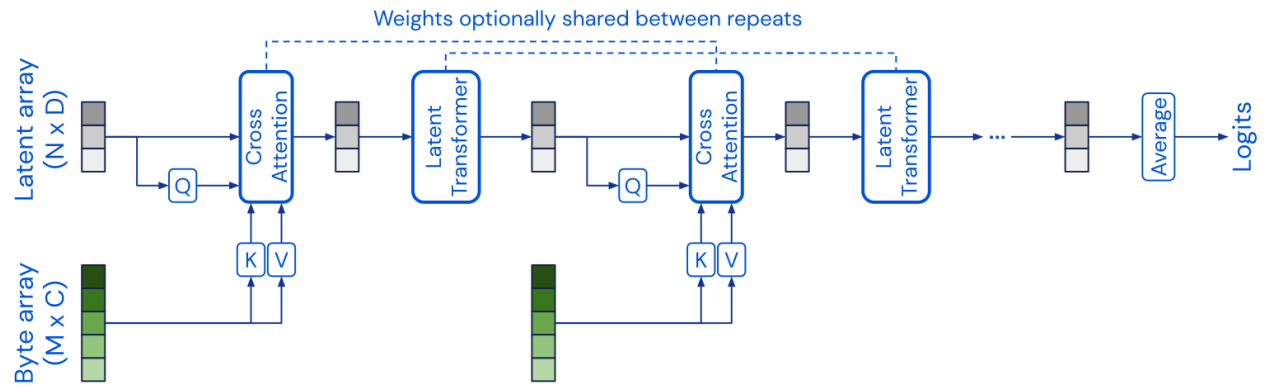


Figure 9: The Perceiver architecture introduces a modality-agnostic framework capable of processing diverse input types, including images, audio, video, and LiDAR. Unlike traditional models that are tailored to specific data structures, the Perceiver imposes minimal assumptions on input format. A key innovation is its use of asymmetric attention, which reduces the quadratic computational complexity of standard transformers, enabling the model to handle inputs with up to 50,000 elements efficiently. This scalability allows it to directly attend to high-resolution sensory inputs that would be computationally prohibitive with conventional transformer models. The Perceiver performed even better than models trained specifically for certain modalities. The flexibility and generality of the Perceiver make it well suited for fusion tasks across heterogeneous sensing domains [23].

9.3 Level 1: OBJECT REFINEMENT Algorithms

These algorithms serve to perform various tasks from object classification to change detection given various data modalities.

9.3.1 Camera

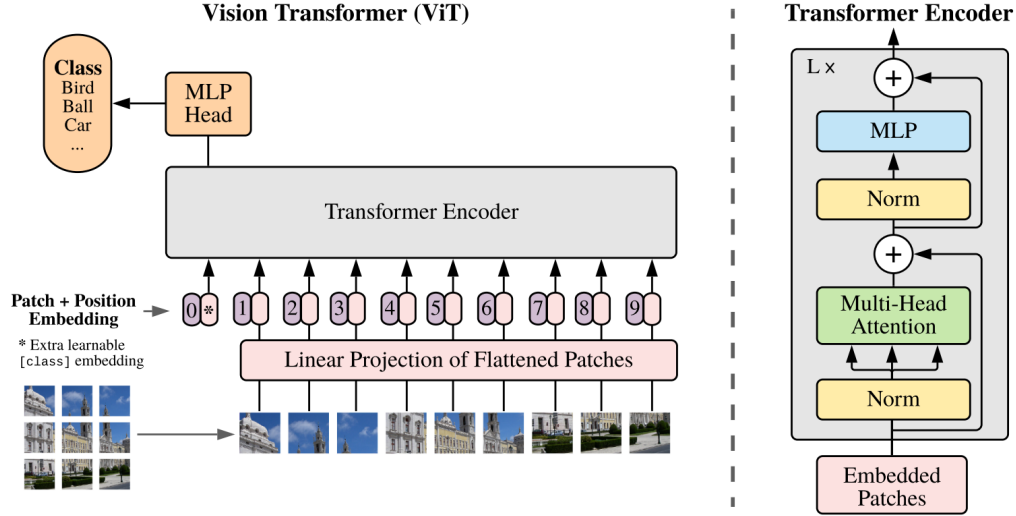


Figure 10: While CNNs have traditionally been the primary tool for image feature extraction, the adoption of attention mechanisms—originally developed for natural language processing—has broadened the scope of image understanding models. This work introduces the Vision Transformer (ViT), which applies the transformer architecture to sequences of image patches. When pretrained on large-scale image-text datasets, ViT matches or exceeds the performance of convolutional models, often with lower computational cost at scale. This architecture demonstrates the feasibility of attention-based models in vision tasks [15].

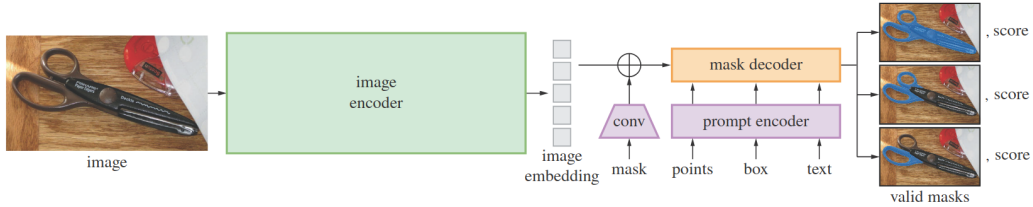


Figure 11: The Segment Anything Model (SAM) introduces a scalable, general-purpose solution for image segmentation. By leveraging a combination of unsupervised learning and curated human-labeled data, the authors construct a large-scale dataset comprising 11 million images and 1.1 billion high-quality segmentation masks. The model is designed as a “foundation model” capable of generalizing to a wide range of segmentation tasks, even with minimal task-specific tuning. However, its reliance on large-scale supervision and image data may limit its flexibility in multi-modal or resource-constrained environments [27].

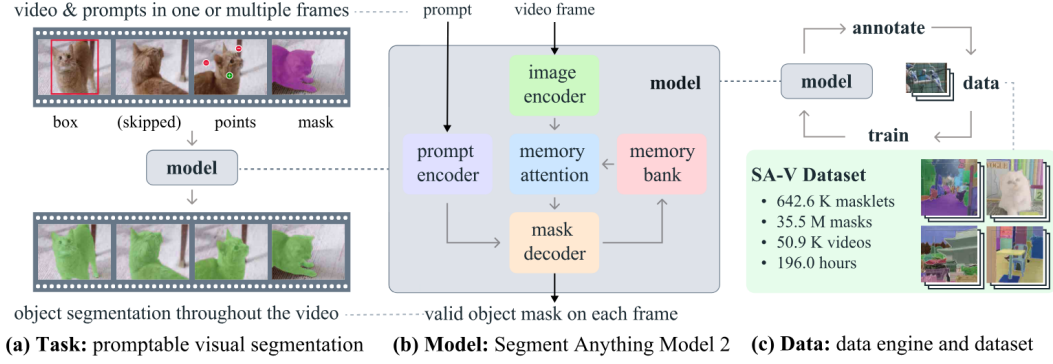


Figure 12: SAM2 builds on the original Segment Anything Model by extending its segmentation capabilities from static images to dynamic video sequences. This includes the ability to track segmented objects across frames, enabling temporal consistency in video understanding. Such functionality is highly relevant for applications in surveillance and activity recognition. Nevertheless, like its predecessor, the model assumes the availability of high-quality visual data and may not easily transfer to non-visual sensing domains [51].

9.3.2 LIDAR

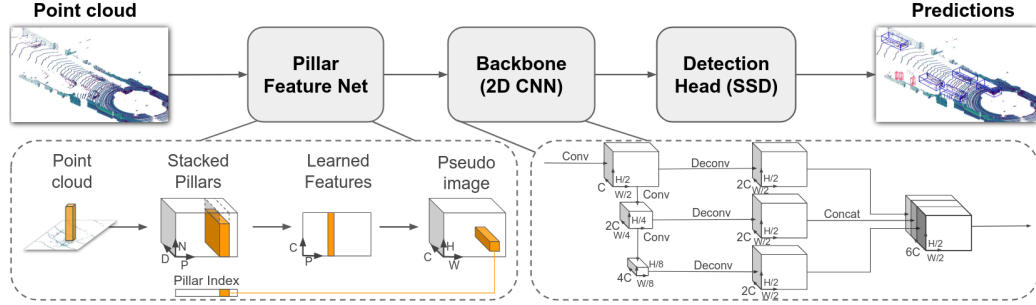


Figure 13: PointPillars addresses the computational burden of 3D object detection in LiDAR point clouds by replacing costly 3D convolutions with a 2D pseudo-image representation. The model divides the input space into vertical columns, or “pillars,” which aggregate point features along the vertical axis, simplifying downstream processing. This design significantly improves inference speed while maintaining competitive accuracy, making it suitable for real-time perception. However, the model’s simplification along the vertical dimension may compromise fine-grained resolution in elevation-sensitive tasks [30].

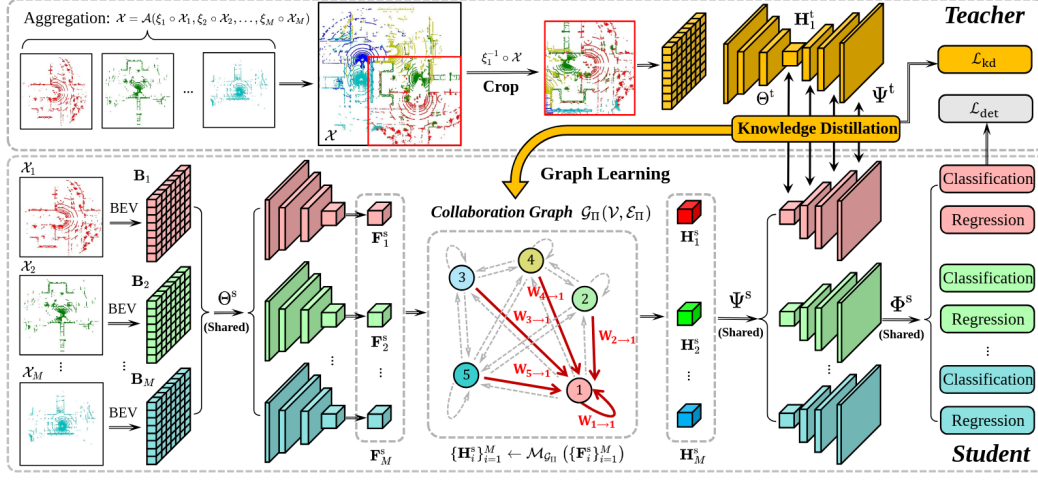


Figure 14: DiscoGraph tackles the challenge of object classification in multi-agent LiDAR perception. It introduces a distilled collaboration graph to determine inter-agent information sharing levels and employs a teacher-student training framework to mitigate communication bandwidth constraints. This approach supports scalable and efficient cooperation among agents. While promising for distributed autonomous systems, its performance depends on assumptions of synchronized data availability and may face limitations in environments with asynchronous or partial observations. The teacher-student framework for handling limited communication constraints might have relevant applicability [33].

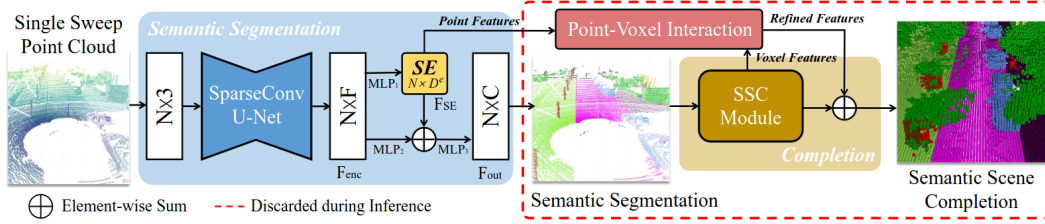


Figure 15: JS3C-Net focuses on enhancing semantic segmentation for sequential LiDAR data. The method captures temporal context by learning spatial priors across multiple scans, improving robustness to occlusions and scene dynamics. This time-aware enhancement is valuable in mobile sensing scenarios. However, its application is limited to sequential data availability and may not generalize well to single-frame or sparsely sampled inputs [65].

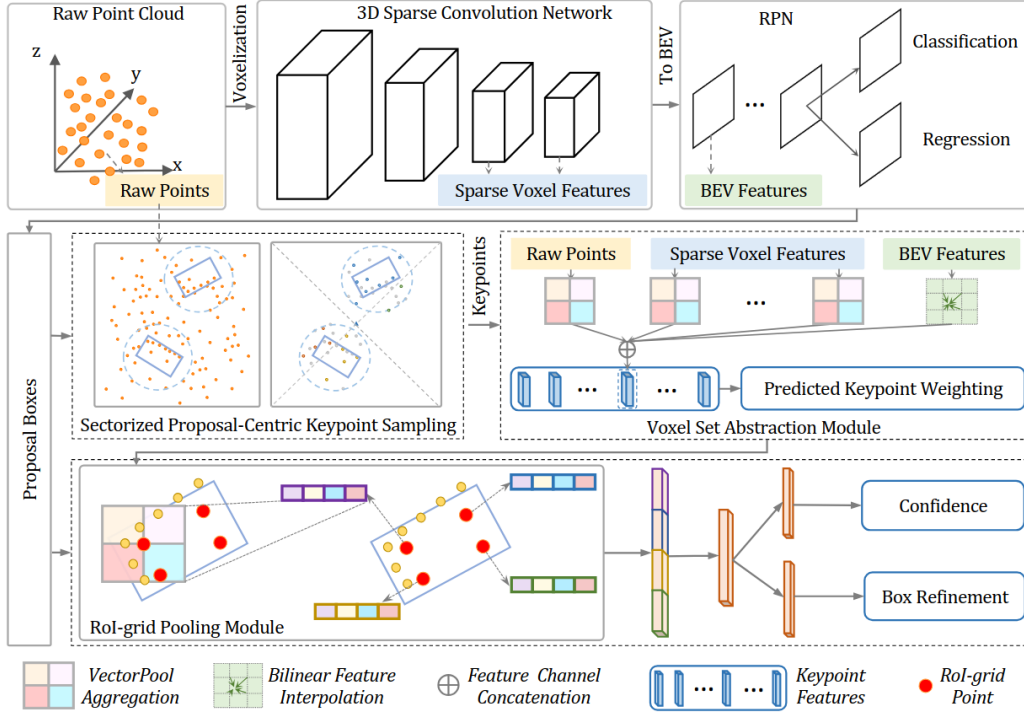


Figure 16: PV-RCNN++ advances LiDAR-based 3D object detection by introducing improved feature sampling and representation techniques, achieving state-of-the-art results on the Waymo Open Dataset in both speed and accuracy. The model enhances the efficiency of local voxel feature encoding and aggregation, supporting accurate detection without incurring high computational costs. While highly effective for autonomous driving benchmarks, its reliance on dense point cloud inputs may hinder performance in sparse or degraded sensing conditions [56].

9.3.3 Radar and Camera

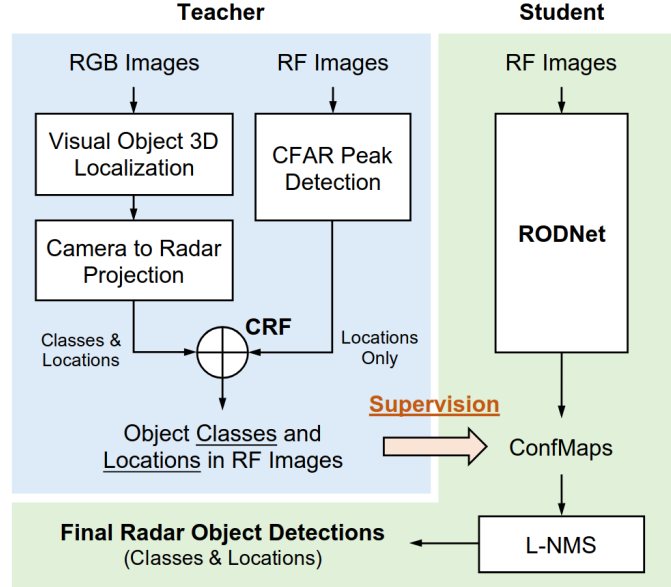


Figure 17: This model leverages a teacher-student training paradigm to transfer knowledge from a richer multimodal sensor system to a more constrained one. The teacher model is trained using both camera and RF-Image data, while the student model relies solely on RF-Image input. During training, the student learns to replicate the teacher’s output, enabling it to approximate the same level of performance with fewer sensory inputs. This approach demonstrates strong potential for cost-efficient deployments in environments where high-resolution or multi-sensor fusion is impractical [64].

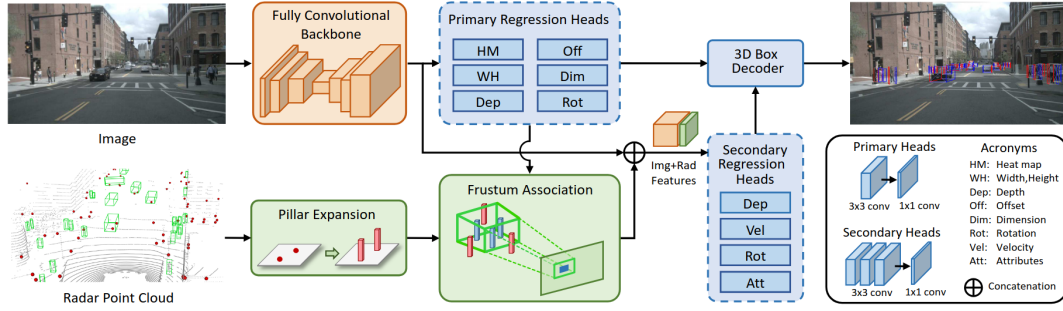


Figure 18: This approach integrates radar point cloud data with visual inputs from a camera to enhance object detection and localization in autonomous driving scenarios. By combining the high spatial resolution of camera imagery with the depth and velocity information from radar, the model improves the placement and precision of 3D bounding boxes. The fusion mechanism is particularly effective in addressing the limitations of single-modality perception systems, especially under low-visibility or adverse weather conditions [41].

9.3.4 SARS and Camera

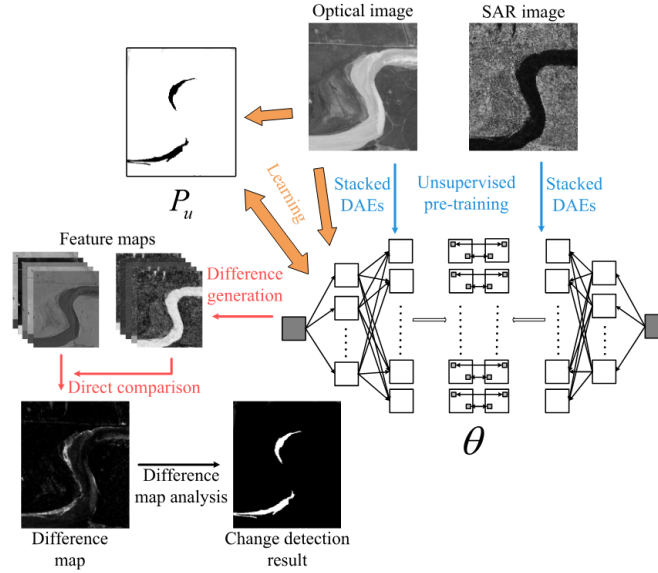


Figure 19: This model performs unsupervised change detection by jointly processing optical camera imagery and synthetic aperture radar (SAR) data. By leveraging the complementary characteristics of both modalities—visual detail from optical images and structural or terrain information from SAR—the model can robustly identify changes in the environment without requiring labeled supervision.

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